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SIXTH SEMESTER **B. Tech. (IT)**

END SEMESTER EXAMINATION ***May-2018***

IT-306 ARTIFICIAL INTELLIGENCE & EXPERT SYSTEMS

Time: 3:00 Hours *Max. Marks: 40*

Note: Answer any *FIVE* questions. Question no. 1 is compulsory
Assume suitable missing data, if any.

1 Answer all the following questions: [2x6=12]

[a] What is beta pruning?

[b] Describe Turing test for intelligence machine. Also, give major characteristics of intelligent machine.

[c] Describe table based and percept based agent architecture.

[d] What is Baye's rule?

[e] What is meant by intelligent program? Explain with example.

[f] What is heuristic search?

2 Answer all the following questions:

[a] Describe Depth First Search (DFS) and compare with Breadth First Search (BFS). [3]

[b] What is meant by state-space search? Explain it using water jug problem. [4]

3 With the help of Tic Tae Toe game, show how Min max procedure help in deciding a good move for maximizing player. [7]

4 Answer all the following questions:

[a] What are the merits and demerits of semantic nets and frames? [3]

[b] Define fuzzy sets and compare with classical sets. Also comment how it is different from probability theory. [4]

5 Answer all the following questions:

[a] Explain Genetic Algorithm with an example. [3]

[b] Draw the architecture of Expert system defining functionality of different components. [4]

6 Answer all the following questions:

[a] Write a note on Learning using Neural networks. [3]

[b] Consider the following sentences:

a. Every child loves dog.
b. All dogs have fur skin or short hair.
c. Roby is a dog
d. Tina does not love dog with short hair
e. Shikha loves Roby
f. Tina does not love dog

(i) Represent theses sentences as wff.
(ii) Using backward chaining prove that Tina is not child.
(iii) Convert them to clause form indicating the law used.
(iv) Explaining steps of resolution, prove that Roby has fur skin. [4]

7 [a] Explain the working of the artificial neural network (ANN) with the help of a neat diagram.

[b] State and explain an evolutionary algorithm for optimal search.

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